ROOT CAUSE ANALYSIS (RCA) REPORT

Objective

The objective of Root Cause Analysis (RCA) is to determine the probable reason behind each anomaly detected in the telecom network.

While anomaly detection identifies when abnormal network behavior occurs, RCA attempts to explain why the anomaly occurred and provides a suitable corrective action for network operators.

Input Data

The RCA module uses the anomalies detected by the Isolation Forest and LSTM anomaly detection models.

The following KPIs are analyzed:

KPI	Description
RSRP	Reference Signal Received Power
SINR	Signal-to-Interference-plus-Noise-Ratio
Latency	Network communication delay
Throughput	Network data transfer rate
Packet Loss	Percentage of packets lost
Connected Users	Number of users connected to a cell

RCA Methodology

A rule-based RCA approach was implemented. Each anomalous record is analyzed against predefined telecom KPI thresholds.

When a KPI exceeds its normal operating range, the system assigns:

Observation ---> Probable Root Cause ---> Recommended Action

For Example :

Low SINR --> High Interference --> Review neighboring cell configuration

Root Cause Rules

Observation	Possible Root Cause	Recommended Action
RSRP<-110 dBm	Poor radio coverage	Optimise antenna tilt

SINR<10 dB	High Interference	Review neighboring cell configuration
Latency>80 ms	Core network congestion	Check UPF utilization
Packet Loss>3%	Backhaul/transport issue	Inspect transport network
Throughput<30 Mbps	Heavy traffic/capacity limitation	Enable load balancing
Connected Users>180	Cell overload	Rebalance users or deploy additional capacity

These thresholds provide an interpretable mechanism for translating abnormal KPI values into potential telecom network problems.

RCA Processing

For every anomaly detected by the anomaly detection model, the RCA system performs the following process:

Detected Anomaly --> Analyze KPI values --> Compare against thresholds --> Identify abnormal KPI --> Assign probable root cause --> Recommend corrective action

For example, suppose an anomalous record contains:

RSRP = -87 dBm
SINR = 7 dB
Latency = 91 ms
Packet Loss = 1.2%
Throughput = 65 Mbps
Connected Users = 120

Two abnormal conditions are identified:

Low SINR
→ High Interference
→ Review neighboring cell configuration

High Latency
→ Core Network Congestion
→ Check UPF utilization

The system therefore associates the anomaly with multiple potential network issues.

RCA Output

The results are stored in: root_cause_analysis.csv

A typical RCA output has the following structure:

Timestamp	Cell ID	Observation	Possible Root Cause	Recommended Action
10:15	Cell_03	Low RSRP	Poor radio coverage	Optimize antenna tilt
11:30	Cell_07	Low SINR	High interference	Review neighboring cells
14:20	Cell_02	High Latency	Core network congestion	Check UPF utilization
17:45	Cell_06	High Packet Loss	Backhaul issue	Inspect transport network
19:10	Cell_04	Low Throughput	Heavy traffic load	Enable load balancing

Root Cause Interpretation

Poor Radio Coverage

Poor radio coverage is associated with very low RSRP values. It may occur because of distance from the base station, physical obstacles, antenna configuration, or insufficient coverage.

Recommended action: Optimize antenna tilt and investigate cell coverage

High Interference

Low SINR indicates that interference or noise is affecting signal quality

Recommended action: Review neighboring cells, frequency allocation, and radio configuration.

Network Congestion

High latency may indicate congestion in the network or excessive utilization of core network resources

Recommended action: Inspect backhaul links and transport infrastructure

Heavy Traffic Load

Low throughput can occur when network capacity is insufficient to handle current traffic

Recommended action: Enable load balancing and investigate capacity expansion

Cell Overload

A very high number of connected users can overload a cell and negatively affect throughput and latency.

Recommended action: Rebalance users across neighboring cells or increase cell capacity.

Benefits of RCA

The RCA module improves the overall AIOps system because it moves beyond simply reporting anomalies. It provides network operators with actionable information.

The major benefits are faster fault investigation, reduced manual troubleshooting, easier prioritization of network problems, improved QoS/QoE, and support for proactive network maintenance.

Limitations

The implemented RCA system is rule-based, meaning that it identifies the probable root cause rather than proving the actual physical cause of a network failure.

For example, high latency can result from core congestion, transport congestion, server delay, or several other factors. Therefore, the RCA result should be treated as a recommendation for further investigation.

A production telecom system would combine KPI information with alarms, topology information, network logs, configuration data, and historical incidents.

Conclusions

The Root Cause Analysis module successfully extends the anomaly detection system by providing an explanation for abnormal telecom network behavior.

Using KPI-based rules, detected anomalies are mapped to probable causes such as poor radio coverage, interference, congestion, backhaul problems, heavy traffic, and cell overload. The system also generates recommended corrective actions for network operators.

Combined with Isolation Forest, LSTM anomaly detection, traffic forecasting, QoS prediction, and the Streamlit dashboard, the RCA module contributes to a complete AIOps-based telecom network monitoring solution.